Digital Signal Processing

Analog and Digital Systems

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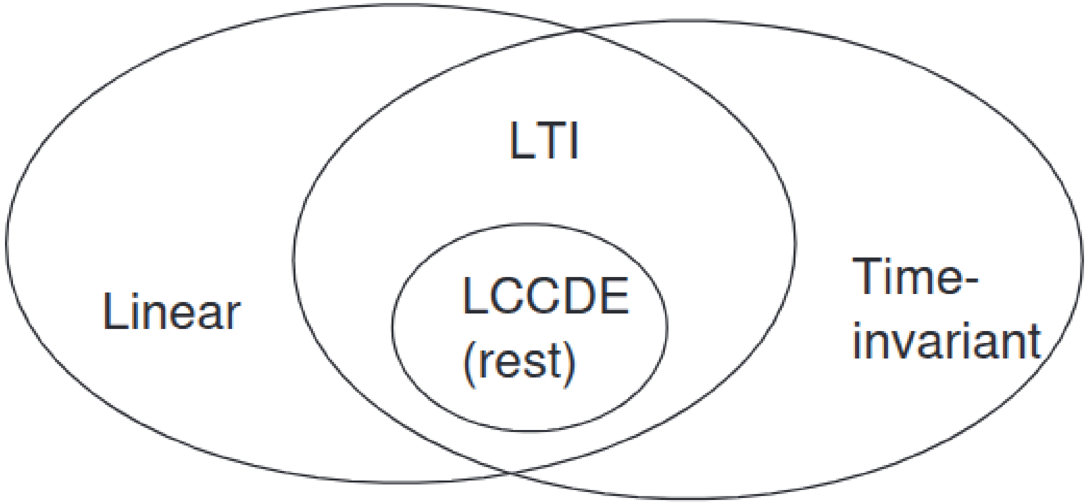
Signals vs Systems

$$x[n] \rightarrow \boxed{\mathcal{H}} \rightarrow y[n] \quad \text{or} \quad x(t) \rightarrow \boxed{\mathcal{H}} \rightarrow y(t)$$

Table 3.1: *Relations of the impulse response to the system function and frequency response of LTI systems.*

Time	System function		Frequency response	
	Nomenclature	Output/input	Nomenclature	Output/input
Continuous	$H(s)$ Laplace of $h(t)$	$H(s) = \frac{Y(s)}{X(s)}$	$H(\omega)$ Fourier of $h(t)$	$H(\omega) = \frac{Y(\omega)}{X(\omega)}$
Discrete	$H(z)$ Z of $h[n]$	$H(z) = \frac{Y(z)}{X(z)}$	$H(e^{j\Omega})$ DTFT of $h[n]$	$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$

Linear Systems



LTI - Linear and Time-Invariant Systems

LCCDE – Linear Constant-Coefficient difference equations

We will focus on LTI systems

In Linear systems, the linear combination of inputs lead to the same linear combination of their respective outputs

x[n] is a sum of scaled shifted impulses

$$x[n] = \sum_{n_0=-\infty}^{\infty} x[n_0]\delta[n-n_0]$$

Using linearity, we obtain that the response to such a signal is the sum of the responses to each scaled shifted impulse

Time-Invariant Systems

By time-invariance, the response to any signal $x[n-k]$ will be $y[n-k]$

If a system is also linear, we only need the response to a single impulse

$$\mathcal{H}\{\delta[n-k]\} = h[n-k]$$

$$y[n] = \mathcal{H}\{x[n]\} = \mathcal{H}\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k]\mathcal{H}\{\delta[n-k]\} \quad \text{(using linearity)}$$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \quad \text{(using time-invariance)}$$

LTI system impulse response example

Assume the impulse response of an LTI system:

$$h[n] = 2\delta[n] + 5\delta[n-1]$$

Now, for the input signal below:

$$x[n] = 4\delta[n] - 3\delta[n-1]$$

We have:

$$\begin{aligned}
y[n] &= \mathcal{H}\{x[n]\} = \mathcal{H}\{4\delta[n] - 3\delta[n-1]\} \\
&= 4\mathcal{H}\{\delta[n]\} - 3\mathcal{H}\{\delta[n-1]\} && \text{(using linearity)} \\
&= 4h[n] - 3h[n-1] && \text{(using time-invariance)} \\
&= 4(2\delta[n] + 5\delta[n-1]) - 3(2\delta[n-1] + 5\delta[n-2]) \\
&= 8\delta[n] + 14\delta[n-1]) - 15\delta[n-2].
\end{aligned}$$

The convolution operator as the core of LTI Systems

$$y[n] = x[n] * h[n]$$

$$y(t) = x(t) * h(t)$$

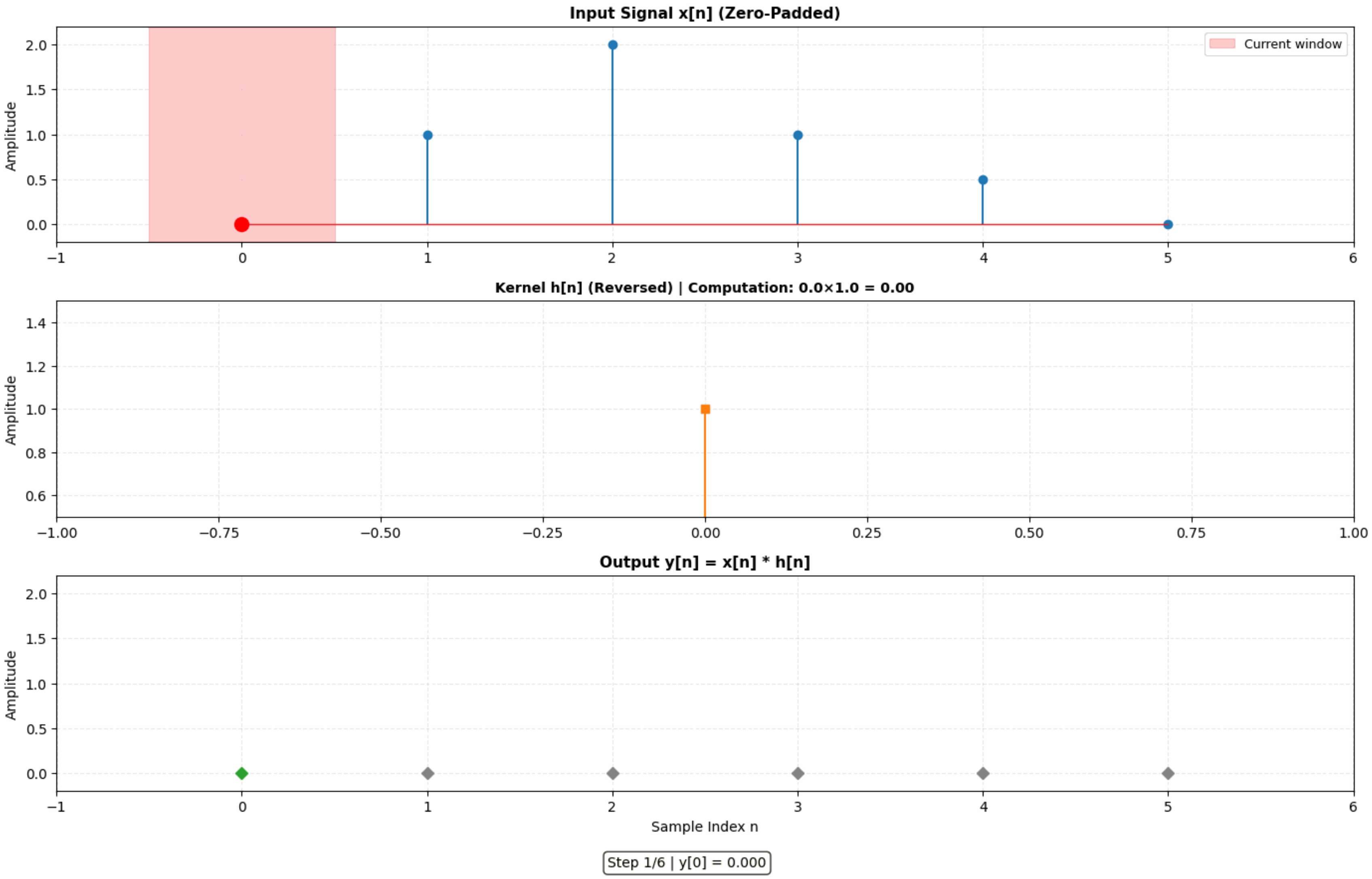
Convolution Formal Expression	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$
Continuous Convolution	$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$	$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

Graphical example of convolution (impulse) - example 1/3

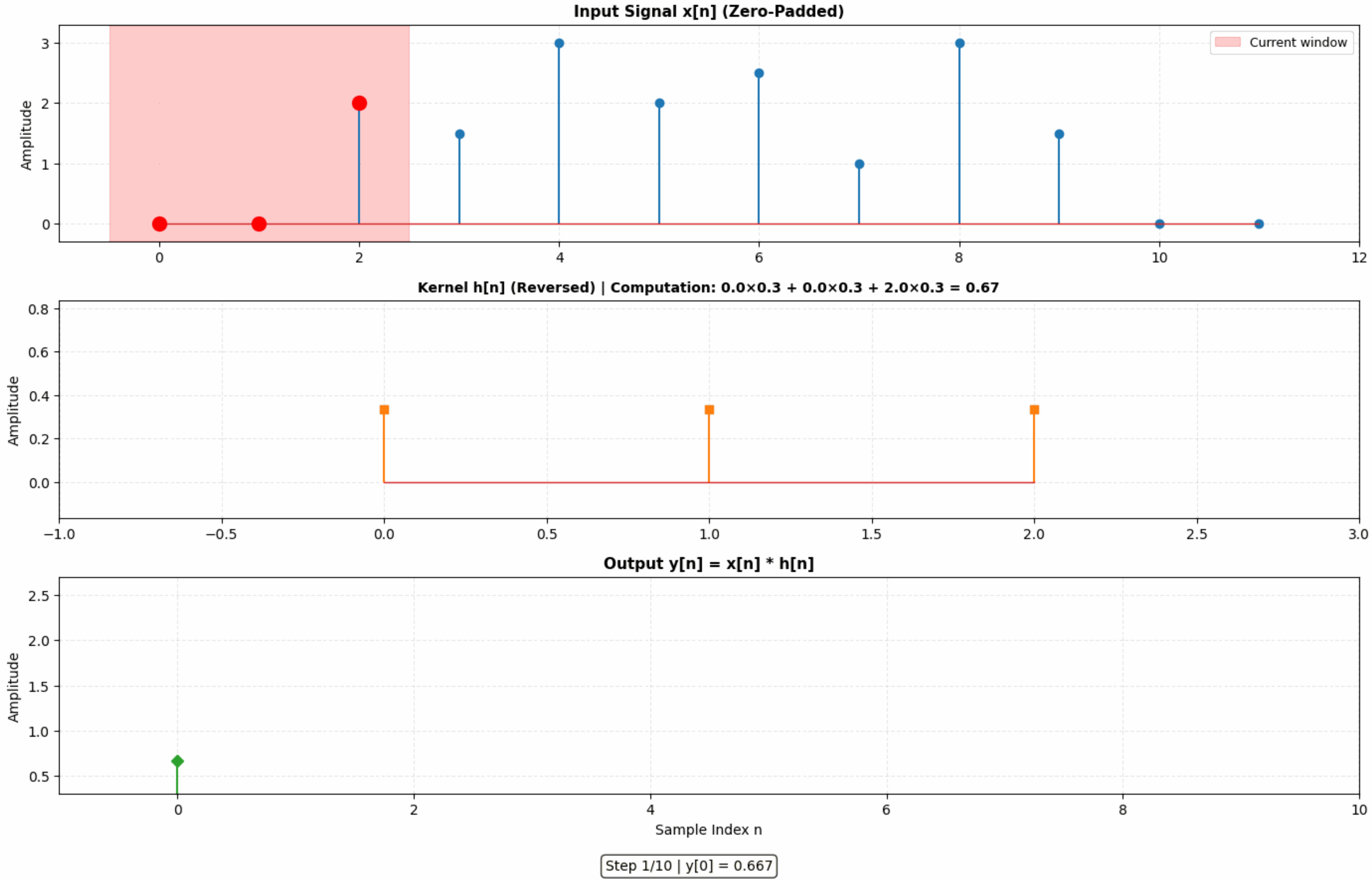
$$x[n]$$

$$h[n] = \delta[n]$$

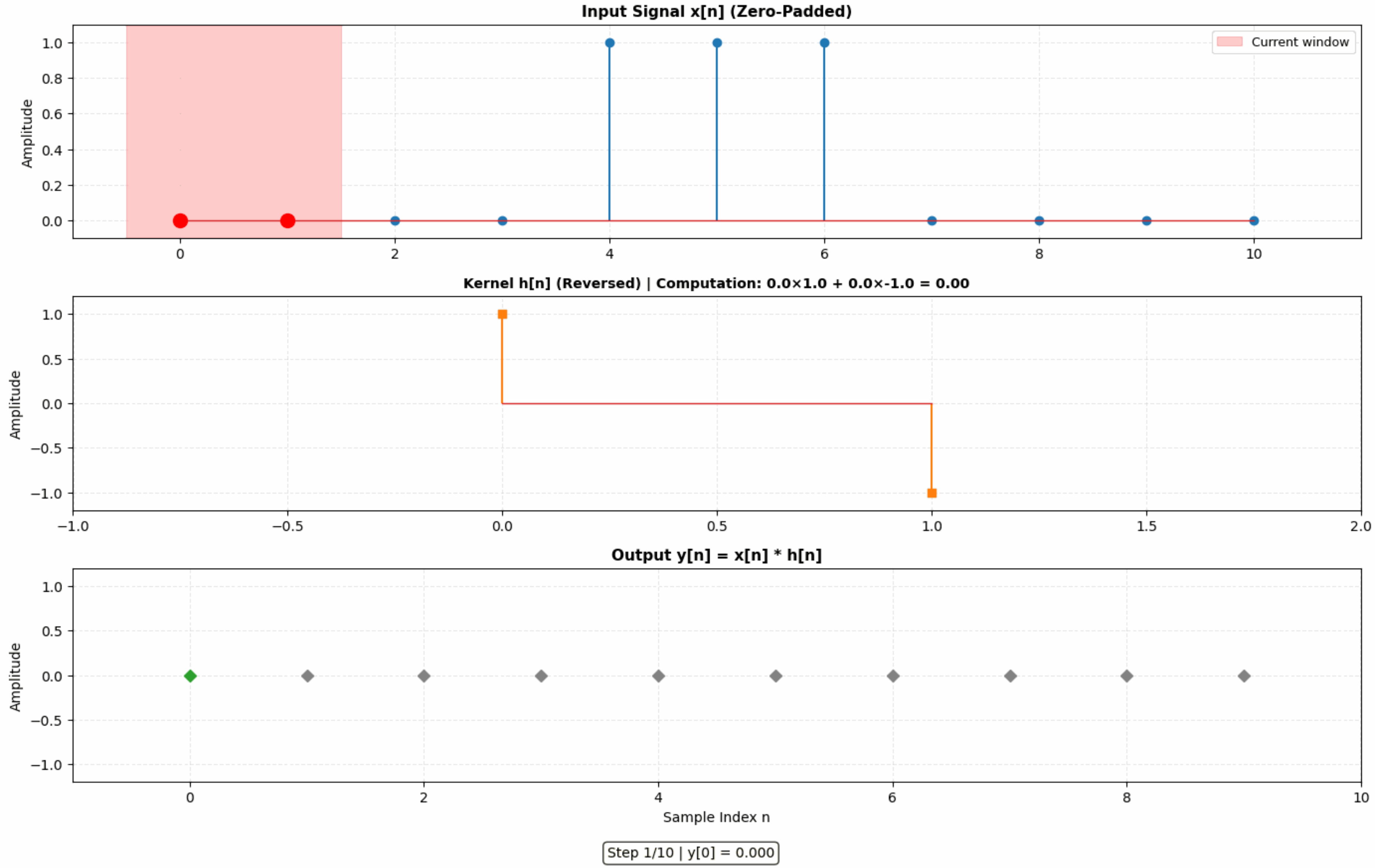
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$



Graphical example of convolution (rectangular) - example 2/3



Graphical example of convolution (edge detector) - example 3/3



Convolution in time domain is a multiplication in frequency domain

$Cx = \lambda x$

For linear operators C, we have eigenvectors and eigenvalues

The LTI system is linear, and it has *eigenfunctions*

$X(e^{j\Omega}) \rightarrow \mathcal{H} \rightarrow Y(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\Omega})$

$x[n] = e^{j\gamma n} \rightarrow \mathcal{H} \rightarrow y[n] = \eta e^{j\gamma n} = |\eta| e^{j\gamma n + \angle\eta}$

In general:

$$x_1[n] * x_2[n] \Leftrightarrow X_1(z)X_2(z)$$

Convolution in time domain is a multiplication in frequency domain

Continuous-time

$$x_1(t)x_2(t) \Leftrightarrow X_1(f)*X_2(f) = \int_{-\infty}^{\infty} X_1(p)X_2(f-p)\mathrm{d}p$$

Discrete-time

$$x_1[n]x_2[n] \Leftrightarrow \frac{1}{2\pi}\int_{<2\pi>} X(e^{j\Omega})X_2(e^{j(\Omega-\theta)})d\theta$$

(Circular) Convolution using DTFTs and inverse DTFT

$$\boxed{x_1[n]x_2[n] \Leftrightarrow \frac{1}{2\pi}\int_{<2\pi>} X(e^{j\Omega})X_2(e^{j(\Omega-\theta)})d\theta}$$

This property allows us to write the above formula as the one below:

$$\boxed{x_1(t)\circledast x_2(t) = \frac{1}{T}\int_{<T>} x_1(\tau)x_2(t-\tau)\mathrm{d}\tau}$$

$$\boxed{x_1[n]x_2[n] \Leftrightarrow \frac{1}{2\pi}\int_{<2\pi>} X(e^{j\Omega})X_2(e^{j(\Omega-\theta)})d\theta} \Leftrightarrow X(e^{j\Omega})\circledast X_2(e^{j\Omega})$$

Circular and fast convolution using FFT

$$y[n] = x_1[n] \circledast x_2[n] = \mathrm{FFT}^{-1}\{X_1[k]X_2[k]\}$$

Suppose $x_1[n]$ has support 4 and $x_2[n]$ has support 3

FFT size = 4, pad with zeroes for x2

1	2	3	4

1	0	-1	0

0	0	0	0	-1	0	1	0	-1	0	1

Circular and fast convolution using FFT

$$y[n] = x_1[n] \circledast x_2[n] = \mathrm{FFT}^{-1}\{X_1[k]X_2[k]\}$$

$x_1[n]$ has support 4 and $x_2[n]$ has support 3

FFT size = 6, pad with zeroes for x1 and x2

1	2	3	4	0	0

1	0	-1	0	0	0

0	0	0	0	0	0	0	0	-1	0	1	0	0	0	-1	0	1

Sampling Theorem

$$\boxed{p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)}$$

$$x_s(t) = x(t)p(t) = \sum_{k=-\infty}^{\infty} x(kT_s)\delta(t - kT_s)$$

$$X_s(f) = X(f) * P(f).$$

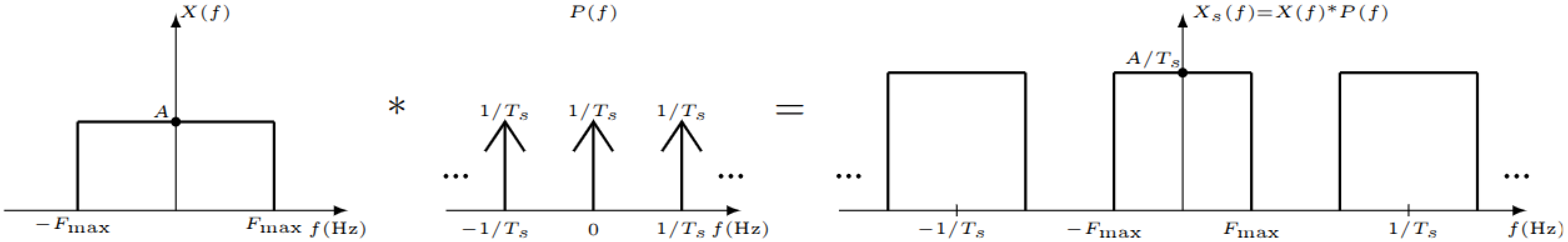
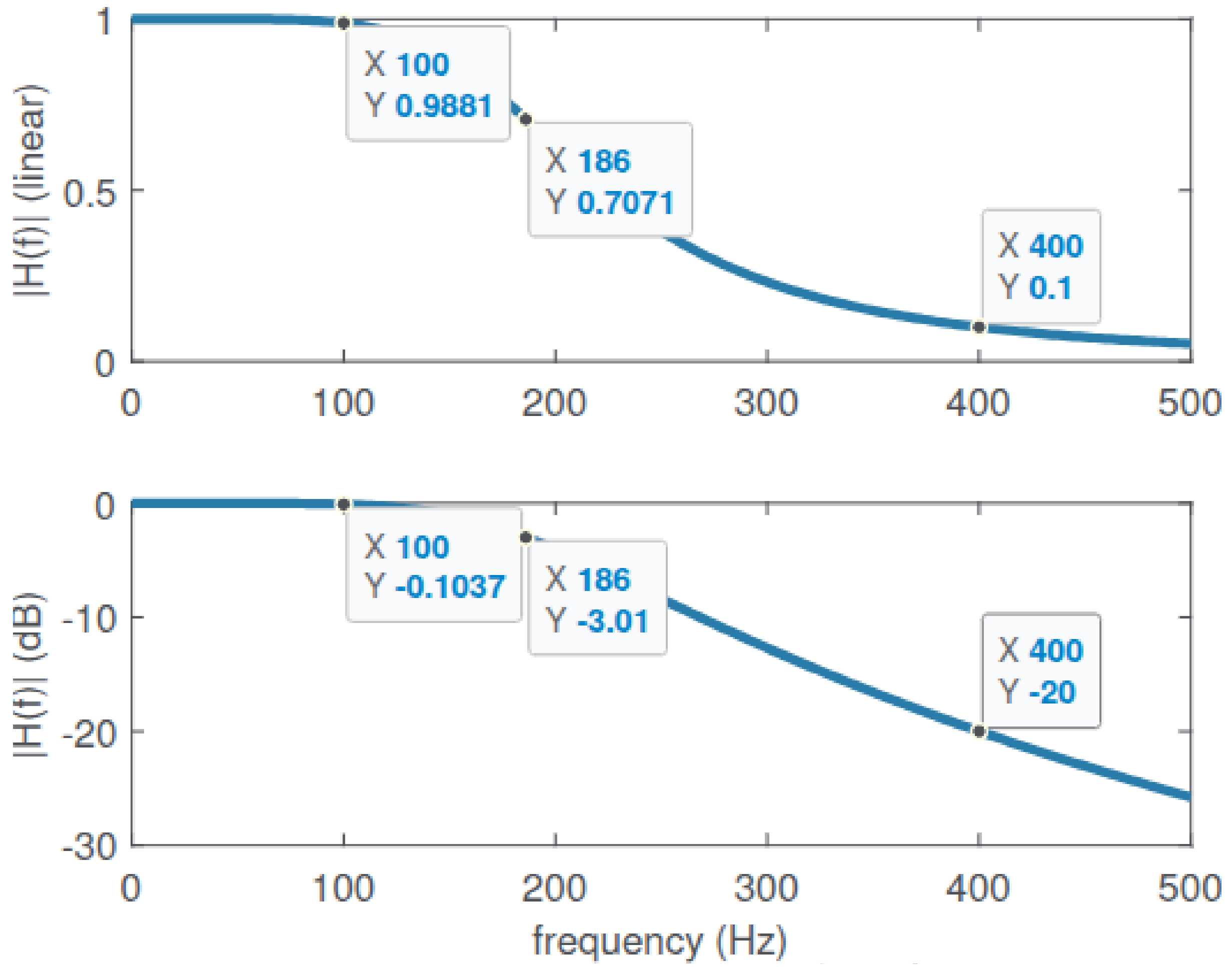


Figure 3.11: *Spectrum $X_s(f)$ of a sampled signal $x_s(t)$ obtained by the convolution between $X(f)$ and $P(f)$ as indicated in Eq. (3.16).*

Discussion on Filters



Symbol	Description
ω_c or f_c	Cutoff frequency: where gain falls by $1/\sqrt{2} \approx 0.707$ (-3 dB)
ω_n	Natural frequency, for example, of a resonator
ω_0	Center frequency of a pole
f_p	Passband frequency
f_r	Stopband (or rejection) frequency
F_s	Sampling frequency (typically in Hz or samples per sec.)
$F_s/2$	Nyquist (or folding) frequency

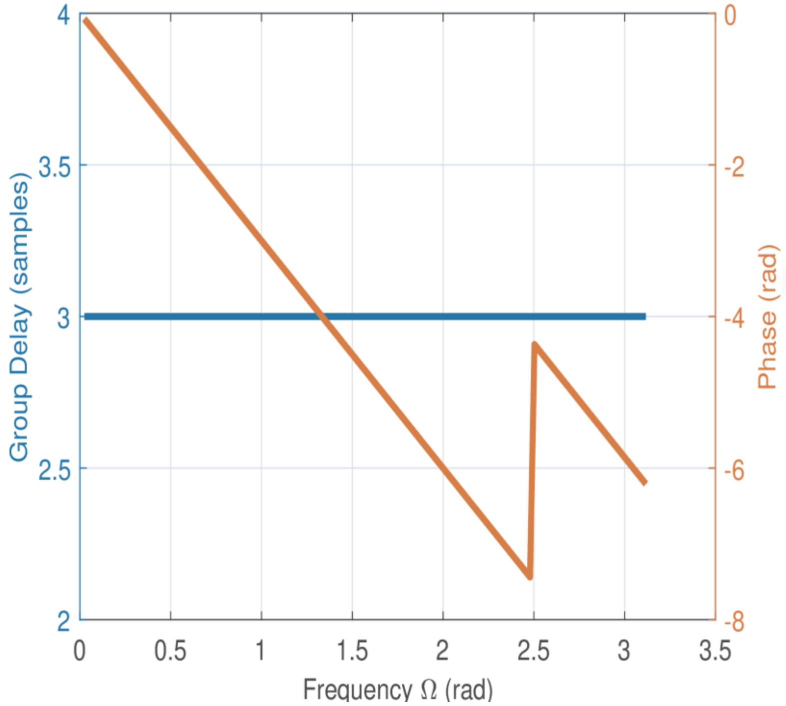
Figure 3.2: *Magnitude of the frequency response of a practical analog filter in linear scale ($|H(f)|$, top) and in dB ($20\log_{10}|H(f)|$), which should be compared to the ideal case of Figure 3.1. Three frequencies are of interest: passband ($f_p = 100$ Hz), cutoff ($f_c = 158.5$ Hz) and stopband ($f_r = 500$ Hz) frequencies.*

Linear Phase Filters

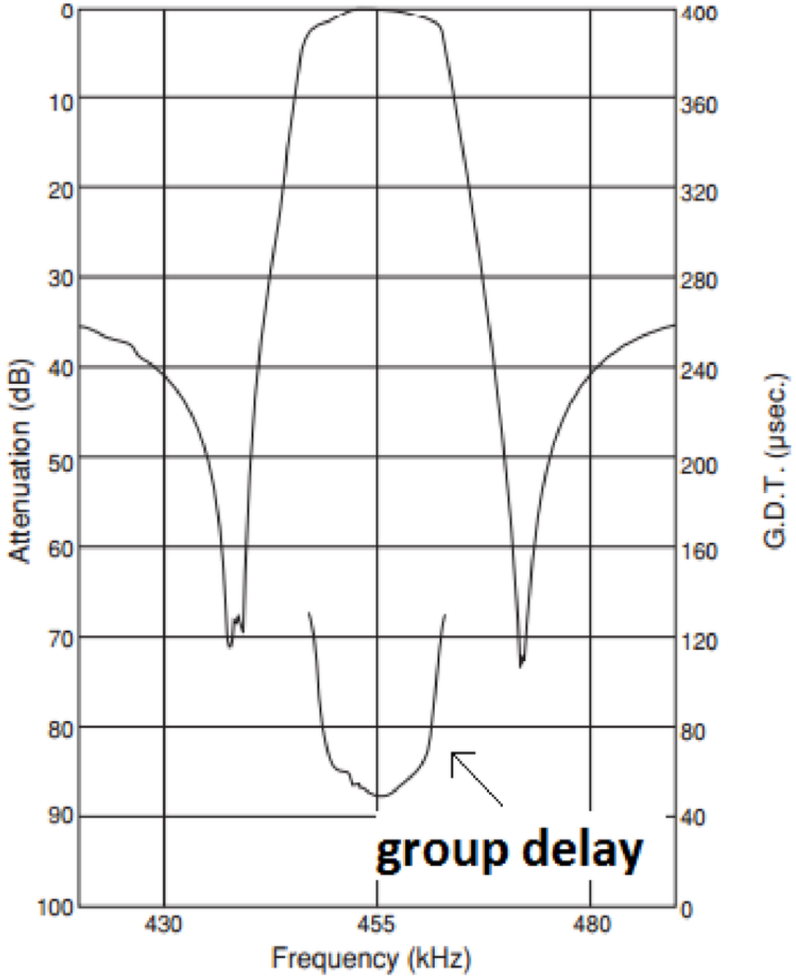
The response delay to a complex exponential is the derivative of the phase of the frequency response to that exponential

$$\tau_g(\omega) = -\frac{d}{d\omega}\angle H(\omega)$$

$$\tau_g\left(e^{j\Omega}\right) = -\frac{d}{d\Omega}\angle H\left(e^{j\Omega}\right)$$



A symmetric FIR filter has constant group delay



Analog filters may get close to but can't achieve linear phase

Thank you!

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